

DETAILED DESCRIPTION

SMBus Interface

The bq24725 operates as a slave, receiving control inputs from the embedded controller host through the SMBus interface. The bq24725 uses a simplified subset of the commands documented in System Management Bus Specification V1.1, which can be downloaded from www.smbus.org. The bq24725 uses the SMBus Read-Word and Write-Word protocols (see [Figure 17](#)) to communicate with the smart battery. The bq24725 performs only as a SMBus slave device with address 0b00010010 (0x12H) and does not initiate communication on the bus. In addition, the bq24725 has two identification registers a 16-bit device ID register (0xFFH) and a 16-bit manufacturer ID register (0xFEH).

SMBus communication is enabled with the following conditions:

- V_{VCC} is above UVLO;
- V_{ACDET} is above 0.6V;

The data (SDA) and clock (SCL) pins have Schmitt-trigger inputs that can accommodate slow edges. Choose pull-up resistors (10kΩ) for SDA and SCL to achieve rise times according to the SMBus specifications. Communication starts when the master signals a START condition, which is a high-to-low transition on SDA, while SCL is high. When the master has finished communicating, the master issues a STOP condition, which is a low-to-high transition on SDA, while SCL is high. The bus is then free for another transmission. [Figure 18](#) and [Figure 19](#) show the timing diagram for signals on the SMBus interface. The address byte, command byte, and data bytes are transmitted between the START and STOP conditions. The SDA state changes only while SCL is low, except for the START and STOP conditions. Data is transmitted in 8-bit bytes and is sampled on the rising edge of SCL. Nine clock cycles are required to transfer each byte in or out of the bq24725 because either the master or the slave acknowledges the receipt of the correct byte during the ninth clock cycle. The bq24725 supports the charger commands as described in [Table 2](#).

a) Write-Word Format

S	SLAVE ADDRESS	W	ACK	COMMAND BYTE	ACK	LOW DATA BYTE	ACK	HIGH DATA BYTE	ACK	P
	7 BITS	1b	1b	8 BITS	1b	8 BITS	1b	8 BITS	1b	
	MSB LSB	0	0	MSB LSB	0	MSB LSB	0	MSB LSB	0	

Preset to 0b0001001 ChargeCurrent() = 0x14H D7 D0 D15 D8
 ChargeVoltage() = 0x15H
 InputCurrent() = 0x3FH
 ChargeOption() = 0x12H

b) Read-Word Format

S	SLAVE ADDRESS	W	ACK	COMMAND BYTE	ACK	S	SLAVE ADDRESS	R	ACK	LOW DATA BYTE	ACK	HIGH DATA BYTE	NACK	P
	7 BITS	1b	1b	8 BITS	1b		7 BITS	1b	1b	8 BITS	1b	8 BITS	1b	
	MSB LSB	0	0	MSB LSB	0		MSB LSB	1	0	MSB LSB	0	MSB LSB	1	

Preset to 0b0001001 DeviceID() = 0xFFH Preset to 0b0001001 D7 D0 D15 D8
 ManufactureID() = 0xFEH
 ChargeCurrent() = 0x14H
 ChargeVoltage() = 0x15H
 InputCurrent() = 0x3FH
 ChargeOption() = 0x12H

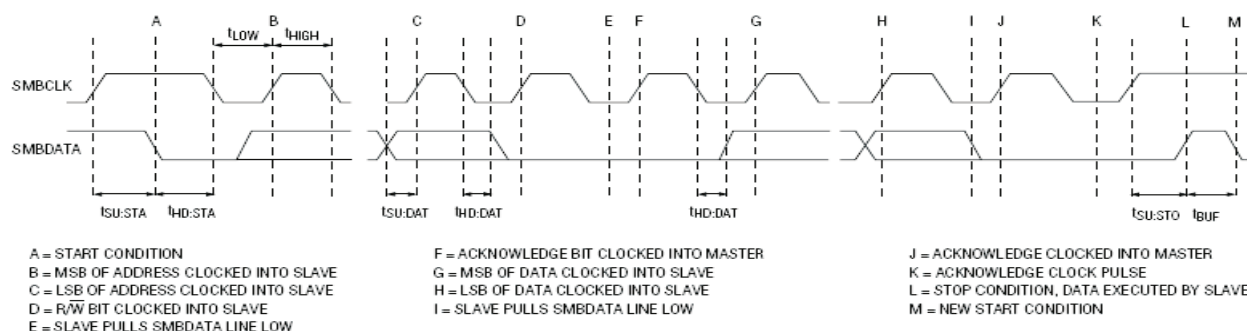
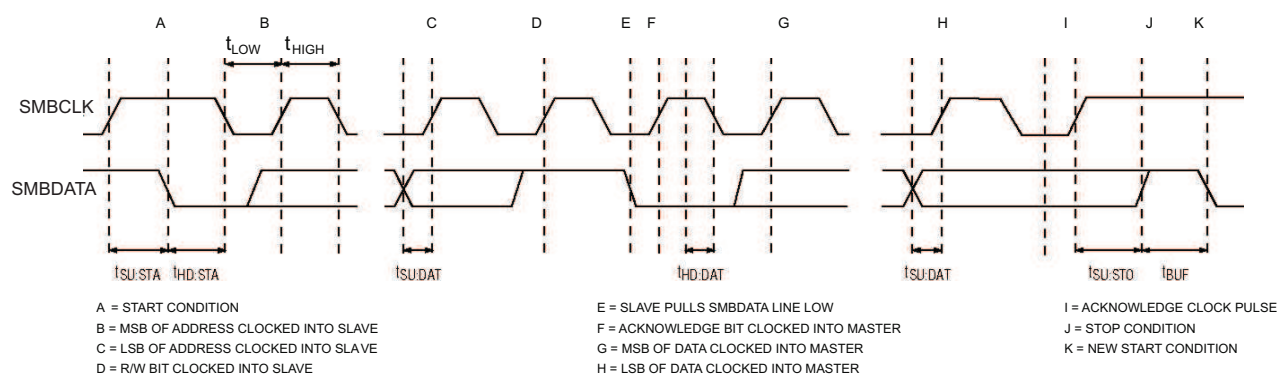
S = START CONDITION OR REPEATED START CONDITION
 ACK = ACKNOWLEDGE (LOGIC-LOW)
 W = WRITE BIT (LOGIC-LOW)

LEGEND:

P = STOP CONDITION
 NACK = NOT ACKNOWLEDGE (LOGIC-HIGH)
 R = READ BIT (LOGIC-HIGH)

	MASTER TO SLAVE
	SLAVE TO MASTER

Figure 17. SMBus Write-Word and Read-Word Protocols


Figure 18. SMBus Write Timing

Figure 19. SMBus Read Timing

Battery-Charger Commands

The bq24725 supports six battery-charger commands that use either Write-Word or Read-Word protocols, as summarized in [Table 2](#). ManufacturerID() and DeviceID() can be used to identify the bq24725. The ManufacturerID() command always returns 0x0040H and the DeviceID() command always returns 0x0008H.

Table 2. Battery Charger Command Summary

REGISTER ADDRESS	REGISTER NAME	READ/WRITE	DESCRIPTION	POR STATE
0x12H	ChargeOption()	Read or Write	Charger Options Control	0x7904H
0x14H	ChargeCurrent()	Read or Write	7-Bit Charge Current Setting	0x0000H
0x15H	ChargeVoltage()	Read or Write	11-Bit Charge Voltage Setting	0x0000H
0x3FH	InputCurrent()	Read or Write	6-Bit Input Current Setting	0x1000H
0XFEH	ManufacturerID()	Read Only	Manufacturer ID	0x0040H
0xFFH	DeviceID()	Read Only	Device ID	0x0008H

Setting Charger Options

By writing ChargeOption() command (0x12H or 0b00010010), bq24725 allows users to change several charger options after POR (Power On Reset) as shown in [Table 3](#).

Table 3. Charge Options Register (0x12H)

BIT	BIT NAME	DESCRIPTION
[15]	ACOK Deglitch Time Adjust	Adjust ACOK deglitch time. To change this option, VCC pin voltage must above UVLO and ACDET pin voltage must above 0.6V to enable IC SMBus communication. After POR the bit default value is 0 and ACOK rising edge deglitch time is 150ms. The long deglitch timer will only be enabled after ACFET turns on and turns off one time this makes first time adapter plug in deglitch time always 150ms 0: ACOK rising edge deglitch time 150ms <default at POR> 1: ACOK rising edge deglitch time 1.3s
[14:13]	WATCHDOG Timer Adjust	Set maximum delay between consecutive SMBus Write charge voltage or charge current command. The charge will be suspended if IC does not receive write charge voltage or write charge current command within the watchdog time period and watchdog timer is enabled. The charge will be resumed after receive write charge voltage or write charge current command when watchdog timer expires and charge suspends. 00: Disable Watchdog Timer 01: Enabled, 44 sec 10: Enabled, 88 sec 11: Enable Watchdog Timer (175s) <default at POR>
[12:11]	BAT Depletion Comparator Threshold Adjust	This is used for LEARN function. During LEARN cycle, when IC detect battery voltage is below depletion voltage threshold IC will turn off BATFET and turn on ACFET to power the system from AC adapter instead of battery. The rising edge hysteresis is typical 305mV to 370mV. 00: Falling Threshold = 59.19% of voltage regulation limit (~2.486V/cell) 01: Falling Threshold = 62.65% of voltage regulation limit (~2.631V/cell) 10: Falling Threshold = 66.55% of voltage regulation limit (~2.795V/cell) 11: Falling Threshold = 70.97% of voltage regulation limit (~2.981V/cell) < default at POR>
[10]	EMI Switching Frequency Adjust	0: Reduce PWM switching frequency by 18% <default at POR> 1: Increase PWM switching frequency by 18%
[9]	EMI Switching Frequency Enable	0: Disable adjust PWM switching frequency <default at POR> 1: Enable adjust PWM switching frequency
[8:7]	IFAULT_HI Comparator Threshold Adjust	Short circuit protection high side MOSFET voltage drop comparator threshold. 00: 300mV 01: 500mV 10: 700mV <default at POR> 11: 900mV
[6]	LEARN Enable	Set this bit 1 start battery learn cycle. IC turns off ACFET and turns on BATFET to discharge battery capacity. When battery voltage reaches threshold defined in bit [12:11], the BATFET is turned off and ACFET is turned on to finish battery learn cycle. After finished learn cycle, this bit is automatically reset to 0. Set this bit 0 will stop battery learn cycle. IC turns off BATFET and turns on ACFET. 0: Disable LEARN Cycle <default at POR> 1: Enable LEARN Cycle
[5]	IOUT Selection	0: IOUT is the 20x adapter current amplifier output <default at POR> 1: IOUT is the 20x charge current amplifier output
[4]	Not In Use	0 at POR
[3]	Not In Use	0 at POR
[2:1]	ACOC Threshold Adjust	00: Disable ACOC 01: 1.33X of input current regulation limit 10: 1.66X of input current regulation limit <default at POR> 11: 2.22X of input current regulation limit
[0]	Charge Inhibit	0: Enable Charge <default at POR> 1: Inhibit Charge

Setting the Charge Current

To set the charge current, write a 16bit ChargeCurrent() command (0x14H or 0b00010100) using the data format listed in [Table 4](#). With 10mΩ sense resistor, the bq24725 provides a charge current range of 128mA to 8.128A, with 64mA step resolution. Sending ChargeCurrent() below 128mA or above 8.128A clears the register and terminates charging. Upon POR, charge current is 0A. A 0.1μF capacitor between SRP and SRN for differential mode filtering is recommended, 0.1μF capacitor between SRN and ground for common mode filtering, and an optional 0.1μF capacitor between SRP and ground for common mode filtering. Meanwhile, the capacitance on SRP should not be higher than 0.1μF in order to properly sense the voltage across SRP and SRN for cycle-by-cycle under-current and over current detection.

The SRP and SRN pins are used to sense R_{SR} with default value of 10mΩ. However, resistors of other values can also be used. For a larger sense resistor, a larger sense voltage is given, and a higher regulation accuracy; but, at the expense of higher conduction loss. If the current sensing resistor value is too high, it may trigger an over current protection threshold because the current ripple voltage is too high. In such a case, either a higher inductance value or a lower current sensing resistor value should be used to limit the current ripple voltage level. A current sensing resistor value no more than 20mΩ is suggested.

To provide secondary protection, the bq24725 has an ILIM pin with which the user can program the maximum allowed charge current. Internal charge current limit is the lower one between the voltage set by ChargeCurrent(), and voltage on ILIM pin. To disable this function, the user can pull ILIM above 1.6V, which is the maximum charge current regulation limit. [Equation 1](#) shows the voltage set on ILIM pin with respect to the preferred charge current limit:

$$V_{ILIM} = 20 \times (V_{SRP} - V_{SRN}) = 20 \times I_{CHG} \times R_{SR} \quad (1)$$

Table 4. Charge Current Register (0x14H), Using 10mΩ Sense Resistor

BIT	BIT NAME	DESCRIPTION
0	–	Not used.
1	–	Not used.
2	–	Not used.
3	–	Not used.
4	–	Not used.
5	–	Not used.
6	Charge Current, DACICHG 0	0 = Adds 0mA of charger current. 1 = Adds 64mA of charger current.
7	Charge Current, DACICHG 1	0 = Adds 0mA of charger current. 1 = Adds 128mA of charger current.
8	Charge Current, DACICHG 2	0 = Adds 0mA of charger current. 1 = Adds 256mA of charger current.
9	Charge Current, DACICHG 3	0 = Adds 0mA of charger current. 1 = Adds 512mA of charger current.
10	Charge Current, DACICHG 4	0 = Adds 0mA of charger current. 1 = Adds 1024mA of charger current.
11	Charge Current, DACICHG 5	0 = Adds 0mA of charger current. 1 = Adds 2048mA of charger current.
12	Charge Current, DACICHG 6	0 = Adds 0mA of charger current. 1 = Adds 4096mA of charger current.
13	–	Not used.
14	–	Not used.
15	–	Not used.

Setting the Charge Voltage

To set the output charge regulation voltage, write a 16bit ChargeVoltage() command (0x15H or 0b00010101) using the data format listed in Table 5. The bq24725 provides charge voltage range from 1.024V to 19.200V, with 16mV step resolution. Sending ChargeVoltage() below 1.024V or above 19.2V clears the register and terminates charging. Upon POR, charge voltage limit is 0V.

The SRN pin is used to sense the battery voltage for voltage regulation and should be connected as close to the battery as possible, and place a decoupling capacitor (0.1μF recommended) as close to IC as possible to the decouple high frequency noise.

Table 5. Charge Voltage Register (0x15H)

BIT	BIT NAME	DESCRIPTION
0	-	Not used.
1	-	Not used.
2	-	Not used.
3	-	Not used.
4	Charge Voltage, DACV 0	0 = Adds 0mV of charger voltage. 1 = Adds 16mV of charger voltage.
5	Charge Voltage, DACV 1	0 = Adds 0mV of charger voltage. 1 = Adds 32mV of charger voltage.
6	Charge Voltage, DACV 2	0 = Adds 0mV of charger voltage. 1 = Adds 64mV of charger voltage.
7	Charge Voltage, DACV 3	0 = Adds 0mV of charger voltage. 1 = Adds 128mV of charger voltage.
8	Charge Voltage, DACV 4	0 = Adds 0mV of charger voltage. 1 = Adds 256mV of charger voltage.
9	Charge Voltage, DACV 5	0 = Adds 0mV of charger voltage. 1 = Adds 512mV of charger voltage.
10	Charge Voltage, DACV 6	0 = Adds 0mV of charger voltage. 1 = Adds 1024mV of charger voltage.
11	Charge Voltage, DACV 7	0 = Adds 0mV of charger voltage. 1 = Adds 2048mV of charger voltage.
12	Charge Voltage, DACV 8	0 = Adds 0mV of charger voltage. 1 = Adds 4096mV of charger voltage.
13	Charge Voltage, DACV 9	0 = Adds 0mV of charger voltage. 1 = Adds 8192mV of charger voltage.
14	Charge Voltage, DACV 10	0 = Adds 0mV of charger voltage. 1 = Adds 16384mV of charger voltage.
15	-	Not used.

Setting Input Current

System current normally fluctuates as portions of the system are powered up or put to sleep. With the input current limit, the output current requirement of the AC wall adapter can be lowered, reducing system cost.

The total input current, from a wall cube or other DC source, is the sum of the system supply current and the current required by the charger. When the input current exceeds the set input current limit, the bq24725 decreases the charge current to provide priority to system load current. As the system current rises, the available charge current drops linearly to zero. Thereafter, all input current goes to system load and input current increases.

During DPM regulation, the total input current is the sum of the device supply current I_{BIAS} , the charger input current, and the system load current I_{LOAD} , and can be estimated as follows:

$$I_{INPUT} = I_{LOAD} + \left[\frac{I_{BATTERY} \times V_{BATTERY}}{V_{IN} \times \eta} \right] + I_{BIAS} \quad (2)$$

where η is the efficiency of the charger buck converter (typically 85% to 95%).

To set the input current limit, write a 16-bit InputCurrent() command (0x3FH or 0b00111111) using the data format listed in Table 6. When using a 10mΩ sense resistor, the bq24725 provides an input-current limit range of 128mA to 8.064A, with 128mA resolution. The suggested input current limit is set to no less than 512mA. Sending InputCurrent() below 128mA or above 8.064A clears the register and terminates charging. Upon POR, the default input current limit is 4096mA.

The ACP and ACN pins are used to sense R_{AC} with default value of 10mΩ. However, resistors of other values can also be used. For a larger sense resistor, larger sense voltage is given, and a higher regulation accuracy; but, at the expense of higher conduction loss.

If input current rises above 108% of input current limit set point, the charger will shut down immediately to allow the input current drop. After charging stops, the charger will soft restart to charge the battery if the adapter still has power left to charge the battery. This prevents a crash if the adapter is overloaded when the system has a high and fast loading transient. The waiting time between shut down and restart charging is a natural response time of the input current limit loop.

Table 6. Input Current Register (0x3FH), Using 10mΩ Sense Resistor

BIT	BIT NAME	DESCRIPTION
0	–	Not used.
1	–	Not used.
2	–	Not used.
3	–	Not used.
4	–	Not used.
5	–	Not used.
6	–	Not used.
7	Input Current, DACIIN 0	0 = Adds 0mA of input current. 1 = Adds 128mA of input current.
8	Input Current, DACIIN 1	0 = Adds 0mA of input current. 1 = Adds 256mA of input current.
9	Input Current, DACIIN 2	0 = Adds 0mA of input current. 1 = Adds 512mA of input current.
10	Input Current, DACIIN 3	0 = Adds 0mA of input current. 1 = Adds 1024mA of input current.
11	Input Current, DACIIN 4	0 = Adds 0mA of input current. 1 = Adds 2048mA of input current.
12	Input Current, DACIIN 5	0 = Adds 0mA of input current. 1 = Adds 4096mA of input current.
13	–	Not used.
14	–	Not used.
15	–	Not used.

Adapter Detect and ACOK Output

The bq24725 uses an ACOK comparator to determine the source of power on VCC pin, either from the battery or adapter. An external resistor voltage divider attenuates the adapter voltage before it goes to ACDET. The adapter detect threshold should typically be programmed to a value greater than the maximum battery voltage, but lower than the maximum allowed adapter voltage.

The open drain ACOK output requires external pull up resistor to system digital rail for a high level. It can be pulled to external rail under the following conditions:

- $V_{VCC} > UVLO$;
- $2.4V < V_{ACDET} < 3.15V$ (not in ACOVP condition, nor in low input voltage condition);
- $V_{VCC} - V_{SRN} > 275mV$ (not in sleep mode);

The rising edge delay default is 150ms after ACDET has a valid voltage to make ACOK pull high. The first time after IC POR always gives a 150ms ACOK rising edge delay no matter what the ChargeOption register value. To change this option, the VCC pin voltage must be above UVLO, and the ACDET pin voltage must be above 0.6V which enables the IC SMBus communication and sets ChargeOption() bit[15] to 1 which sets the ACOK rising deglitch time to be 1.3s. Only after the ACDET pin voltage is pulled below 2.4V (but not below 0.6V which will reset IC and force the ACOK rising edge deglitch time to be 150ms) and ACDRV has been turned off at least one time, the 1.3s delay time is effective for next time the ACDET pin voltage goes above 2.4V. The purpose of the option 1.3s rising edge deglitch time is to turn off the ACFET long enough when the ACDET pin is pulled below 2.4V by excessive system current, such as over current or short circuit.

Adapter Over Voltage (ACOV)

When the ACDET pin voltage is higher than 3.15V, it is considered as adapter over voltage. ACOK will be pulled low, and charge will be disabled. ACFET will be turned off to disconnect the high voltage adapter to system during ACOV. BATFET will be turned on if turns on conditions are valid. See the [System Power Selection](#) section for details.

When ACDET pin voltage falls below 3.15V and above 2.4V, it is considered as adapter voltage returns back to normal voltage. ACOK will be pulled high by external pull up resistor. BATFET will be turned off and ACFET and RBFET will be turned on to power the system from adapter. The charge can be resumed if enable charge conditions are valid. See the [Enable and Disable Charging](#) section for details.

System Power Selection

The bq24725 automatically switches adapter or battery power to system. The battery is connected to system at POR if battery exists. The battery is disconnected from system and the adapter is connected to system after default 150ms delay (can be change to 1.3s delay) if ACOK goes HIGH. An automatic break-before-make logic prevents shoot-through currents when the selectors switch.

The ACDRV drives a pair of common-source (CMSRC) n-channel power MOSFETs (ACFET and RBFET) between adapter and ACP (see [Figure 1](#) for details). The ACFET separates adapter from battery or system, and provides a limited di/dt when plugging in adapter by controlling the ACFET turn-on time. Meanwhile it protects adapter when system or battery is shorted. The RBFET provides negative input voltage protection and battery discharge protection when adapter is shorted to ground, and minimizes system power dissipation with its low $R_{DS(on)}$ compared to a Schottky diode.

When the adapter is not present, ACDRV is pulled to CMSRC to keep ACFET and RBFET off, disconnecting adapter from system. BATDRV stays at $V_{SRN} + 6V$ to connect battery to system if all the following conditions are valid:

- $V_{VCC} > UVLO$;
- $V_{SRN} > UVLO$;
- $V_{ACN} < 200mV$ above V_{SRN} (ACN_SRN comparator);

Approximately 150ms (default, can be changed to 1.3s) after the adapter is detected (ACDET pin voltage between 2.4V and 3.15V), the system power source begins to switch from battery to adapter if all the following conditions are valid:

- Not in LEARN mode or in LEARN mode and V_{SRN} is lower than battery depletion threshold;
- ACOK high

The gate drive voltage on ACFET and RBFET is $V_{CMSRC} + 6V$. If the ACFET/RBFET have been turned on for 20ms, and the voltage across gate and source is still less than 5.9V, ACFET and RBFET will be turned off. After 1.3s delay, it resumes turning on ACFET and RBFET. If such a failure is detected seven times within 90 seconds, ACFET/RBFET will be latched off and an adapter removal and system shut down is required to force ACDET < 0.6V to reset the IC. After IC reset from latch off, ACFET/RBFET can be turned on again. After 90 seconds, the failure counter will be reset to zero to prevent latch off. With ACFET/RBFET off, charge is disabled.

To turn off ACFET/RBFET, one of the following conditions must be valid:

- In LEARN mode and V_{SRN} is above battery depletion threshold;
- ACOK low

To limit the in-rush current on ACDRV pin, CMSRC pin and BATDRV pin, a 4kΩ resistor is recommended on each of the three pins.

To limit the adapter inrush current when ACFET is turned on to power system from adapter, the Cgs and Cgd external capacitor of ACFET must be carefully selected. The larger the Cgs and Cgd capacitance, the slower turn on of ACFET will be and less inrush current of adapter. However, if Cgs or Cgd is too large, the ACDRV-CMSRC voltage may still go low after the 20ms turn on time window is expired. To make sure ACFET will not be turned on when adapter is hot plugged in, the Cgs value should be 20 times or higher than Cgd. The most cost effective way to reduce adapter in-rush current is to minimize system total capacitance.

Battery LEARN Cycle

A battery LEARN cycle can be activated via SMBus command (ChargeOption() bit[6]=1 enable LEARN cycle, bit[6]=0 disable LEARN cycle). When LEARN is enabled with ACFET/RBFET connected, the system power selector logic is over-driven to switch to battery by turning off ACFET/RBFET and turning on BATFET. LEARN function allows the battery to discharge in order to calibrate the battery gas gauge over a complete discharge/charge cycle. The controller automatically exits LEARN cycle when the battery voltage is below battery depletion threshold, and the system switches back to adapter input by turning off BATFET and turning on ACFET/RBFET. After LEARN cycle, the LEARN bit is automatically reset to 0. The battery depletion threshold can be set to 59.19%, 62.65%, 66.55%, and 70.97% of voltage regulation level via SMBus command (ChargeOption() bit[12:11]).

Enable and Disable Charging

In Charge mode, the following conditions have to be valid to start charge:

- Charge is enabled via SMBus (ChargeOption() bit [0]=0, default is 0, charge enabled);
- ILIM pin voltage higher than 105mV;
- All three regulation limit DACs have valid value programmed;
- ACOK is valid (See the [Adapter Detect and ACOK Output](#) section for details);
- ACFET and RBFET turns on and gate voltage is high enough (See the [System Power Selection](#) section for details);
- V_{SRN} does not exceed BATOVP threshold;
- IC Temperature doesn't exceed TSHUT threshold;
- Not in ACOC condition (See the [Input Over Current Protection \(ACOC\)](#) section for details);

One of the following conditions will stop on-going charging:

- Charge is inhibited via SMBus (ChargeOption() bit[0]=1);
- ILIM pin voltage lower than 75mV;
- One of three regulation limit DACs is set to 0 or out of range;
- ACOK is pulled low (See the [Adapter Detect and ACOK Output](#) section for details);
- ACFET turns off;
- V_{SRN} exceeds BATOVP threshold;
- TSHUT IC temperature threshold is reached;
- ACOC is detected (See the [Input Over Current Protection \(ACOC\)](#) section for details);
- Short circuit is detected (See the [Inductor Short, MOSFET Short Protection](#) section for details);
- Watchdog timer expires if watchdog timer is enabled (See the [Charger Timeout](#) section for details);

Automatic Internal Soft-Start Charger Current

Every time the charge is enabled, the charger automatically applies soft-start on charge current to avoid any overshoot or stress on the output capacitors or the power converter. The charge current starts at 128mA, and the step size is 64mA in CCM mode for a 10mΩ current sensing resistor. Each step lasts around 240μs in CCM mode, till it reaches the programmed charge current limit. No external components are needed for this function. During DCM mode, the soft start up current step size is larger and each step lasts for longer time period due to the intrinsic slow response of DCM mode.

High Accuracy Current Sense Amplifier

As an industry standard, high accuracy current sense amplifier (CSA) is used to monitor the input current or the charge current, selectable via SMBUS (ChargeOption() bit[5]=0 select the input current, bit[5]=1 select the charge current) by host. The CSA senses voltage across the sense resistor by a factor of 20 through the IOUT pin. Once VCC is above UVLO and ACDET is above 0.6V, CSA turns on and IOUT output becomes valid. If the user wants to lower the voltage on current monitoring, then use a resistor divider from IOUT to GND, and still achieve accuracy over temperature.

A 100pF capacitor connected on the output is recommended for decoupling high-frequency noise. An additional RC filter is optional, if additional filtering is desired. Note that adding filtering also adds additional response delay.

Charge Timeout

The bq24725 includes a watchdog timer to terminate charging if the charger does not receive a write ChargeVoltage() or write ChargeCurrent() command within 175s (adjustable via ChargeOption() command). If a watchdog timeout occurs all register values keep unchanged but charge is suspended. Write ChargeVoltage() or write ChargeCurrent() commands must be re-sent to reset watchdog timer and resume charging. The watchdog timer can be disabled, or set to 44s, 88s or 175s via SMBus command (ChargeOption() bit[14:13]). After watchdog timeout write ChargeOption() bit[14:13] to disable watchdog timer also resume charging.

Converter Operation

The synchronous buck PWM converter uses a fixed frequency voltage mode control scheme and internal type III compensation network. The LC output filter gives a characteristic resonant frequency

$$f_o = \frac{1}{2\pi \sqrt{L_o C_o}} \quad (3)$$

The resonant frequency f_o is used to determine the compensation to ensure there is sufficient phase margin and gain margin for the target bandwidth. The LC output filter should be selected to give a resonant frequency of 10–20 kHz nominal for the best performance. Suggest component value as charge current of 750kHz default switching frequency is shown in Table 7.

Ceramic capacitors show a dc-bias effect. This effect reduces the effective capacitance when a dc-bias voltage is applied across a ceramic capacitor, as on the output capacitor of a charger. The effect may lead to a significant capacitance drop, especially for high output voltages and small capacitor packages. See the manufacturer's data sheet about the performance with a dc bias voltage applied. It may be necessary to choose a higher voltage rating or nominal capacitance value in order to get the required value at the operating point.

Table 7. Suggest Component Value as Charge Current of Default 750kHz Switching Frequency

Charge Current	2A	3A	4A	6A	8A
Output Inductor L_o (μ H)	6.8 or 8.2	5.6 or 6.8	3.3 or 4.7	3.3	2.2
Output Capacitor C_o (μ F)	20	20	20	30	40
Sense Resistor ($m\Omega$)	10	10	10	10	10

The bq24725 has three loops of regulation: input current, charge current and charge voltage. The three loops are brought together internally at the error amplifier. The maximum voltage of the three loops appears at the output of the error amplifier EAO. An internal saw-tooth ramp is compared to the internal error control signal EAO (see Figure 16) to vary the duty-cycle of the converter. The ramp has offset of 200mV in order to allow 0% duty-cycle.

When the battery charge voltage approaches the input voltage, EAO signal is allowed to exceed the saw-tooth ramp peak in order to get a 100% duty-cycle. If voltage across BTST and PHASE pins falls below 4.3V, a refresh cycle starts and low-side n-channel power MOSFET is turned on to recharge the BTST capacitor. It can achieve duty cycle of up to 99.5%.

Continuous Conduction Mode (CCM)

With sufficient charge current the bq24725's inductor current never crosses zero, which is defined as continuous conduction mode. The controller starts a new cycle with ramp coming up from 200mV. As long as EAO voltage is above the ramp voltage, the high-side MOSFET (HSFET) stays on. When the ramp voltage exceeds EAO voltage, HSFET turns off and low-side MOSFET (LSFET) turns on. At the end of the cycle, ramp gets reset and LSFET turns off, ready for the next cycle. There is always break-before-make logic during transition to prevent cross-conduction and shoot-through. During the dead time when both MOSFETs are off, the body-diode of the low-side power MOSFET conducts the inductor current.

During CCM mode, the inductor current is always flowing and creates a fixed two-pole system. Having the LSFET turn-on keeps the power dissipation low, and allows safely charging at high currents.

Discontinuous Conduction Mode (DCM)

During the HSFET off time when LSFET is on, the inductor current decreases. If the current goes to zero, the converter enters Discontinuous Conduction Mode. Every cycle, when the voltage across SRP and SRN falls below 5mV (0.5A on 10mΩ), the under current-protection comparator (UCP) turns off LSFET to avoid negative inductor current, which may boost the system via the body diode of HSFET.

During the DCM mode the loop response automatically changes. It changes to a single pole system and the pole is proportional to the load current.

Both CCM and DCM are synchronous operation with LSFET turn-on every clock cycle. If the average charge current goes below 125mA on 10mΩ current sensing resistor or the battery voltage falls below 2.5V, the LSFET keeps turn-off. The battery charger operates in non-synchronous mode and the current flows through the LSFET body diode. During non-synchronous operation, the LSFET turns on only for a refreshing pulse to charge the BTST capacitor. If the average charge current goes above 250mA on 10mΩ current sensing resistor, the LSFET exits non-synchronous mode and enters synchronous mode to reduce LSFET power loss.

Input Over Current Protection (ACOC)

The bq24725 cannot maintain the input current level if the charge current has been already reduced to zero. After the system current continues increasing to the 1.66X of input current DAC set point (with 2.5ms blank out time), ACFET/RBFET is turned off and charge is disabled for 1.3s and will soft start again for charge if the ACOC condition goes away. If such a failure is detected seven times in 90 seconds, the ACFET and RBFET latch off and an adapter removal and system shutdown is required to force ACDET < 0.6V to reset IC. After IC reset from latch off, ACFET/RBFET can be turned on again. After 90 seconds, the failure counter will be reset to zero to prevent latch off.

The ACOC function can be disabled or the threshold can be set to 1.33X, 1.66X or 2.22X of input DPM current via SMBus command (ChargeOption() bit [2:1]).

Charge Over Current Protection (CHGOC)

The bq24725 has a cycle-by-cycle peak over current protection. It monitors the voltage across SRP and SRN, and prevents the current from exceeding of the threshold based on the DAC charge current set point. The high-side gate drive turns off for the rest of the cycle when the over current is detected, and resumes when the next cycle starts.

The charge OCP threshold is automatically set to 6A, 9A, and 12A on a 10mΩ current sensing resistor based on charge current register value. This prevents the threshold to be too high which is not safe or too low which can be triggered in normal operation. Proper inductance should be selected to prevent OCP triggered in normal operation due to high inductor current ripple.

Battery Over Voltage Protection (BATOV)

The bq24725 will not allow the high-side and low-side FET to turn-on when the battery voltage at SRN exceeds 104% of the regulation voltage set-point. If BATOV last over 30ms, charger is completely disabled. This allows quick response to an over voltage condition – such as occurs when the load is removed or the battery is disconnected. A 4mA current sink from SRN to GND is on only during BATOV and allows discharging the stored output inductor energy that is transferred to the output capacitors.

Battery Shorted to Ground (BATLOWV)

The bq24725 will disable charge for 1ms if the battery voltage on SRN falls below 2.5V. After 1ms reset, the charge is resumed with soft-start if all the enable conditions in the [Enable and Disable Charging](#) sections are satisfied. This prevents any overshoot current in inductor which can saturate inductor and may damage the MOSFET. The charge current is limited to 0.5A on 10mΩ current sensing resistor when BATLOWV condition persists and LSFET keeps off. The LSFET turns on only for a refreshing pulse to charge the BTST capacitor.

Thermal Shutdown Protection (TSHUT)

The QFN package has low thermal impedance, which provides good thermal conduction from the silicon to the ambient, to keep junctions temperatures low. As added level of protection, the charger converter turns off for self-protection whenever the junction temperature exceeds the 155°C. The charger stays off until the junction temperature falls below 135°C. During thermal shut down, the REGN LDO current limit is reduced to 16mA. Once the temperature falls below 135°C, charge can be resumed with soft start.

EMI Switching Frequency Adjust

The charger switching frequency can be adjusted $\pm 18\%$ to solve EMI issue via SMBus command. ChargeOption() bit [9]=0 disable the frequency adjust function. To enable frequency adjust function, set ChargeOption() bit[9]=1. Set ChargeOption() bit [10]=0 to reduce switching frequency, set bit[10]=1 to increase switching frequency.

If frequency is reduced, for a fixed inductor the current ripple is increased. The inductor value must be carefully selected, so that it will not trigger a cycle-by-cycle peak over current protection even for the worst condition such as higher input voltage, 50% duty cycle, lower inductance and lower switching frequency.

Inductor Short, MOSFET Short Protection

The bq24725 has a unique short circuit protection feature. Its cycle-by-cycle current monitoring feature is achieved through monitoring the voltage drop across $R_{DS(on)}$ of the MOSFETs after a certain amount of blanking time. In case of MOSFET short or inductor short circuit, the over current condition is sensed by two comparators and two counters will be triggered. After seven times of short circuit events, the charger will be latched off and ACFET and RBFET is turned off to disconnect adapter from system. BATFET is turned on to connect battery pack to system. To reset the charger from latch-off status, the IC VCC pin must be pulled below UVLO or the ACDET pin must be pulled below 0.6V. This can be achieved by removing the adapter and shut down the operation system. The low side MOSFET short circuit voltage drop threshold is fixed to typical 110mV. The high side MOSFET short circuit voltage drop threshold can be adjusted via SMBus command. ChargeOption() bit[8:7] = 00, 01, 10, 11 set the threshold 300mV, 500mV, 700mV and 900mV respectively.

Due to the certain amount of blanking time to prevent noise when MOSFET just turns on, the cycle-by-cycle charge over current protection may detect high current and turn off MOSFET first before the short circuit protection circuit can detect short condition because the blanking time has not finished. In such a case the charge may not be able to detect shorts circuit and counter may not be able to count to seven then latch off. Instead the charge may continuously keep switching with very narrow duty cycle to limit the cycle-by-cycle current peak value. However, the charger should still be safe and will not cause failure because the duty cycle is limited to a very short of time and MOSFET should be still inside the safety operation area. During a soft start period, it may takes long time instead of just seven switching cycles to detect short circuit based on the same blanking time reason.

Table 8. Component List for Typical System Circuit of [Figure 1](#)

PART DESIGNATOR	QTY	DESCRIPTION
C1, C2, C3, C13, C14, C16	6	Capacitor, Ceramic, 0.1μF, 25V, 10%, X7R, 0603
C4	1	Capacitor, Ceramic, 100pF, 25V, 10%, X7R, 0603
C5, C6	2	Capacitor, Ceramic, 1μF, 25V, 10%, X7R, 0603
C7	1	Capacitor, Ceramic, 0.047μF, 25V, 10%, X7R, 0603
C8, C9, C10, C11	4	Capacitor, Ceramic, 10μF, 25V, 10%, X7R, 1206
C15	1	Capacitor, Ceramic, 0.01μF, 25V, 10%, X7R, 0603
C17	1	Capacitor, Ceramic, 2200pF, 25V, 10%, X7R, 0603
Ci	1	Capacitor, Ceramic, 2.2μF, 25V, 10%, X7R, 1210

Table 8. Component List for Typical System Circuit of Figure 1 (continued)

PART DESIGNATOR	QTY	DESCRIPTION
Csys	1	Capacitor, Electrolytic, 220μF, 25V
D1	1	Diode, Schottky, 30V, 200mA, SOT-23, Fairchild, BAT54
D2	1	Diode, Dual Schottky, 30V, 200mA, SOT-23, Fairchild, BAT54C
Q1, Q2, Q5	3	N-channel MOSFET, 30V, 12.5A, SO-8, Fairchild, FDS6680A
Q3, Q4	2	N-channel MOSFET, 30V, 12A, PowerPAK 1212-8, Vishay Siliconix, SiS412DN
Q6	1	N-channel MOSFET, 50V, 0.2A, SOT-323, Diodes, BSS138W
L1	1	Inductor, SMT, 4.7μH, 5.5A, Vishay Dale, IHLP2525CZER4R7M01
R1	1	Resistor, Chip, 430kΩ, 1/10W, 1%, 0603
R2	1	Resistor, Chip, 66.5kΩ, 1/10W, 1%, 0603
R3, R4, R5	3	Resistor, Chip, 10kΩ, 1/10W, 1%, 0603
R6, R10, R11	3	Resistor, Chip, 4.02kΩ, 1/10W, 1%, 0603
R7	1	Resistor, Chip, 316kΩ, 1/10W, 1%, 0603
R8	1	Resistor, Chip, 100kΩ, 1/10W, 1%, 0603
R9	1	Resistor, Chip, 10Ω, 1/4W, 1%, 1206
R12	1	Resistor, Chip, 1.00MΩ, 1/10W, 1%, 0603
R13	1	Resistor, Chip, 3.01MΩ, 1/10W, 1%, 0603
R14	1	Resistor, Chip, 10Ω, 1/10W, 5%, 0603
R15	1	Resistor, Chip, 7.5Ω, 1/10W, 5%, 0603
RAC, RSR	2	Resistor, Chip, 0.01Ω, 1/2W, 1%, 1206
Ri	1	Resistor, Chip, 2Ω, 1/2W, 1%, 1210
U1	1	Charger controller, 20 pin VQFN, TI, bq24725RGR
U2	1	Dual digital transistor, 40V, 30mA, SC-74, Rohm, IMD2A

APPLICATION INFORMATION

Negative Output Voltage Protection

Reversely insert the battery pack into the charger output during production or hard shorts on battery to ground will generate negative output voltage on SRP and SRN pin. IC internal electrostatic-discharge (ESD) diodes from GND pin to SRP or SRN pins and two anti-parallel (AP) diodes between SRP and SRN pins can be forward biased and negative current can pass through the ESD diodes and AP diodes when output has negative voltage. Insert two resistors for SRP and SRN pins to limit the negative current level when output has negative voltage. Suggest resistor value is 10Ω for SRP pin and 7-8Ω for SRN pin. After adding small resistors, the suggested pre-charge current is at least 192mA for a 10mΩ current sensing resistor.

Reverse Input Voltage Protection

Q6, R12 and R13 in Figure 1 gives system and IC protection from reversed adapter voltage. In normal operation, Q6 is turned off by negative Vgs. When adapter voltage is reversed, Q6 Vgs is positive. As a result, Q6 turns on to short gate and source of Q2 so that Q2 is off. Q2 body diode blocks negative voltage to system. However, CMSRC and ACDRV pins need R10 and R11 to limit the current due to the ESD diode of these pins when turned on. Q6 must has low Vgs threshold voltage and low Qgs gate charge so it turns on before Q2 turns on. R10 and R11 must have enough power rating for the power dissipation when the ESD diode is on. In Figure 2, the Schottky diode D3 gives the reverse adapter voltage protection, no extra small MOSFET and resistors are needed.

Reduce Battery Quiescent Current

When the adapter is not present, if VCC is powered with voltage higher than UVLO directly or indirectly (such as through a LDO or switching converter) from battery, the internal BATFET charge pump gives the BATFET pin 6V higher voltage than the SRN pin to drive the n-channel BATFET. As a result, the battery has higher quiescent current. This is only necessary when the battery powers the system due to a high system current that goes through the MOSFET channel instead of the body diode to reduce conduction loss and extend the battery